How Market Competition Affects Privacy Choices: Competition between Information Collectors

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Abstract

This paper studies how competition between platforms affects the privacy protection they offer to users. We build a two-period model in which a user's activity on a platform reveals information about her type, and the platform chooses a protection level that adds noise to what it learns. The information is valuable for the platform and costly for the user, and it only matters on the platform where the user makes her transactions. An incumbent platform serves the user in the first period, and in the second period a new platform without any data can compete for her. We use the user's switching cost to measure how strong competition is. Our main result is that privacy protection is not monotone in the switching cost. When switching is cheap, the user protects her data the most, because the new platform's offer depends on how much she told the incumbent. For intermediate switching costs, the incumbent collects less data on purpose in order to keep the user. For higher switching costs, the user is "insured" by her outside option and she shares the most data. We also discuss welfare and some policies, such as lowering switching costs and the right to erase data.

1 Introduction

Privacy is becoming an important issue for digital platforms. When we search, click, watch or buy things online, platforms learn about us from our own activity. They can use this information to make money, for example by selling targeted ads or by helping sellers to set prices. Many people worry that platforms collect too much data, and a popular argument is that this happens because there is not enough competition. For example, Srinivasan (2019) argues that Facebook promised not to track its users when it was competing with MySpace, and that it started to track them after it became dominant (see also Economides and Lianos, 2021). Following this idea, policy reports suggest lowering switching costs through interoperability and data portability, and privacy laws like the GDPR give users some rights over their data (Stigler Committee on Digital

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Platforms, 2019; Crémer et al., 2019). However, from a theory point of view it is not clear how the intensity of competition affects the privacy protection of platforms when data is collected over time.

In this paper, we build a simple dynamic model to study this question. The model has two main features. First, data is persistent. What a platform learns today, it can still use tomorrow. A forward-looking user knows this, so she is careful with her data, and how careful she is depends on how her data will be used in the future. Second, data is platform specific and it is used where the user makes transactions. If the user switches to a new platform, the new platform knows nothing about her, but the old platform cannot "forget" what it has learned. Because of this, the incumbent's data is a liability when it competes for the user: the more it knows, the worse is the best deal it can offer to keep her.

In the model, a user with a normally distributed type interacts with platforms for two periods. In each period she chooses an activity level, which generates a noisy signal about her type, and the platform chooses a protection level that adds noise to this signal, following Ichihashi (2023). The information is valuable for the platform and costly for the user, for example because sellers on the platform use it for price discrimination. An incumbent serves the user in the first period. In the second period, a new platform (the rival) without any data can compete for her, and the user pays a switching cost $s \geq 0$ if she moves to the rival. The switching cost measures how strong competition is: $s = 0$ is a fully contestable market and $s = \infty$ is a monopoly.

We first look at a static benchmark. Protection has two effects on what the platform learns. It adds noise to each unit of activity, but it also makes the user more active. Because of the second effect, even a monopoly platform chooses some protection, and we call this the *privacy Laffer curve*. In our Gaussian setting, the monopoly platform always induces the same activity level and always learns the same share $\tilde{x}$ of the remaining uncertainty about the user, no matter how much it already knows (Proposition 1). This property makes the dynamic model easy to solve.

The dynamic part of the model works through the user's continuation payoff. We show that it is the maximum of three linear functions of her first-period disclosure x_1:

$$W(x_1) \;=\; \max\left\{\, v(\tilde{a}) - c\,(\tilde{x} + \kappa x_1),\;\; v(M) - \max\{s,\; c\,x_1\} \,\right\},$$

where c is the user's cost of full disclosure and $\kappa = 1 - \tilde{x}$ (Lemma 2). The slope of W is what we call the user's *exposure* $e \in \{\kappa, 0, 1\}$, that is, how much one unit of data given today costs her tomorrow. When the incumbent just wants to collect as much data as possible in the first period, the user discloses $x_1 = \tilde{x}/\sqrt{1 + \beta_u e}$, where β_u is her discount factor (Lemma 3). So market structure affects privacy in two ways: through the user's exposure (the user side) and through the incumbent's incentive to keep the user (the platform side).

Our main result, Theorem 1, shows that privacy in the first period is not monotone in the switching cost. There are five regimes:

- *Poaching* (low s). The rival wins the user in period 2, but it has to beat the best offer of the incumbent, and this offer is worse when the incumbent knows more. So the

rival's offer is tied ("pegged") to the incumbent's data, and the user's exposure is one. She is more careful than under monopoly, so privacy in the first period is higher, even though the incumbent knows that it will lose the user.

- *Limit collection* (moderately low s). The incumbent can keep the user if its data liability is not larger than the switching cost. So it collects less data in period 1 on purpose, exactly up to this level. This retention motive only exists with competition, and in this regime privacy decreases with the switching cost.
- *Insured incumbency* (moderately high s). The rival's offer, minus the switching cost, fixes the user's future payoff, and this does not depend on her history. The user is "insured", her exposure is zero and she discloses the most. The incumbent only offers the static monopoly protection.
- *Transition and monopoly* (high s). When the insurance becomes less valuable, the incumbent has to add protection to keep the user insured. For even higher s this is not possible anymore, and we get the monopoly outcome with exposure κ.

So privacy is highest when switching costs are low but positive, lowest when they are moderately high, and in between under monopoly.

These results also have some welfare implications (Proposition 3). The user is best off with a small but positive switching cost, namely the smallest one at which the incumbent prefers to keep her rather than to lose her. For high but finite switching costs, she is worse off than under monopoly: she is insured by a weak outside option, so she stops protecting her data and discloses more. On the other side, the incumbent prefers some competition to a full lock-in. A credible rival commits the incumbent not to exploit the user later, and this makes the user more willing to be active today. This is similar to the second-sourcing idea of Farrell and Gallini (1988).

We also discuss some policies. Lowering switching costs, for example with interoperability or data portability, improves privacy only if the switching cost becomes low enough that the incumbent is afraid of losing the user. A small reduction from a high level can hurt users. A right to erase data removes the user's exposure, so she stops protecting her data and discloses more. She is still better off when switching costs are high, because protecting herself was costly. But the right can hurt her when switching costs are low, mainly because the incumbent then has no reason to limit its data collection (Proposition 4). Finally, when platforms compete for the market before any data is collected and the winner then locks the user in, privacy is complete in the first period and then goes down. This is a privacy version of "bargains then rip-offs", and it fits the Facebook story above (Proposition 5).

The rest of the paper is organized as follows. Section 2 reviews the literature. Section 3 presents the model. Section 4 studies the static and the monopoly benchmarks. Section 5 contains the main analysis with competition. Section 6 discusses welfare and policy, Section 7 discusses some extensions, and Section 8 concludes. All proofs are in Appendix B.

2 Literature Review

This paper is related to three groups of papers. The first group studies dynamic privacy choices. Our paper is most related to Ichihashi (2023), and we use his framework of activity and protection. He studies a monopoly platform and a consumer in an infinite horizon. With long-run commitment, the platform lowers protection over time. With one-period commitment, there can be an equilibrium in which protection goes down over time and also an equilibrium without any data collection, depending on the consumer's expectations. Our monopoly benchmark has a similar pattern of decreasing protection, in a two-period model with continuous activity, where the equilibrium is unique and in closed form. The main difference is that we study competition, which is not in his paper. The behavior of the user in our model is similar to the ratchet effect (Freixas et al., 1985; Laffont and Tirole, 1988; Hart and Tirole, 1988): she is less active today because what she reveals today will be used against her tomorrow. Similar forward-looking behavior appears when firms use past behavior to set prices or scores (Taylor, 2004; Acquisti and Varian, 2005; Conitzer et al., 2012; Bonatti and Cisternas, 2020). In labor markets, competition between employers reduces the ratchet effect, because the future wage of a worker is decided by the market (Kanemoto and MacLeod, 1992; Charness et al., 2011). In our model the answer depends on the form of competition. When the rival's offer fixes the user's future payoff, competition removes the user's caution, but when the rival's offer depends on the incumbent's data, competition makes her even more careful.

The second group studies competition and privacy. Casadesus-Masanell and Hervas-Drane (2015) study firms that compete for consumers and also sell consumer information in a secondary market. Kox et al. (2017) study competing ad-funded platforms that choose how much to track users. Choi et al. (2019) show that information externalities lead to too much data collection. Ichihashi (2021) studies competition between data intermediaries, and Bergemann and Bonatti (2024) and de Cornière and Taylor (2025) provide frameworks to study how data affects competition. Jullien et al. (2020) study a website that trades off selling data against keeping consumers who learn from bad experiences, and Fainmesser et al. (2023) study data collection and data protection when stored data can be leaked. About regulation, Campbell et al. (2015) show that consent-based privacy rules can favor larger firms, which is consistent with the higher concentration after the GDPR found by Johnson et al. (2023). See Acquisti et al. (2016) and Goldfarb and Que (2023) for surveys. Compared with these papers, we focus on a mechanism that is specific to data. Since data is persistent and platform specific, the incumbent's data is a liability in competition, and the user's exposure depends on the market structure. This leads to a non-monotone relationship between competition and privacy.

The third group studies switching costs and customer poaching. The second period of our model is a Bertrand game with switching costs (Klemperer, 1987, 1995; Farrell and Klemperer, 2007), where old and new customers are treated differently, as in the literature on customer recognition and poaching (Villas-Boas, 1999; Fudenberg and Tirole, 2000; Taylor, 2003; Fudenberg and Villas-Boas, 2006). There are two differences.

First, the "price" in our model is privacy, and the incumbent's information about the customer is a liability and not an asset. Second, when competition happens before lock-in, we get a privacy version of the "bargains then rip-offs" pattern. Our result that the incumbent may prefer some competition is similar to second sourcing and licensing as commitment devices (Shepard, 1987; Farrell and Gallini, 1988). Lam and Liu (2020) study whether data portability helps entry.

3 The Model

3.1 Setup

Players and information. There is a user (she) with type $X \sim \mathcal{N}(\mu, \sigma_0^2)$, and she knows her type. There are two periods, $t \in \{1, 2\}$. An incumbent platform A is available in both periods, and a rival platform E becomes available in period 2. In each period the user uses exactly one of the available platforms and chooses an *activity level* $a_t \in [0, M]$ on it, for example how much she browses, posts or searches. Activity gives her enjoyment $v(a_t)$, with $v(0) = 0$.

Before the user chooses her activity, the platform she uses commits to a *protection level* $\gamma \in [0, \infty]$. The platform then observes a signal

$$z_t = X + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}\left(0,\ \gamma + \tfrac{1}{a_t}\right), \tag{1}$$

where the noise is independent of X and across periods. If $a_t = 0$ or $\gamma = \infty$, the signal has no information. The precision of the signal, $a_t/(1 + a_t\gamma)$, is increasing in activity but it is never larger than $1/\gamma$. So protection puts an upper bound on how much the platform can learn in one period, which is similar to the idea of differential privacy.

Information stock. Let $\sigma_{i,t}^2$ be the posterior variance of platform i about X at the end of period t, and let

$$y_{i,t} \ := \ 1 - \sigma_{i,t}^2/\sigma_0^2 \ \in [0, 1]$$

be its *information stock*, which is the share of the prior variance that it has learned. By Bayes' rule, if a platform with stock y serves the user and she chooses activity a under protection γ, the platform learns the share

$$\xi(a, \gamma; y) \ = \ \frac{a\,\sigma^2}{1 + a\,(\sigma^2 + \gamma)}, \qquad \sigma^2 := \sigma_0^2(1 - y), \tag{2}$$

of its remaining uncertainty, and its stock becomes $y + (1 - y)\,\xi(a, \gamma; y)$. A platform keeps what it has learned, and the stock of a platform that the user does not use stays the same. We write $x_1 := y_{A,1} = \xi(a_1, \gamma_1; 0)$ for the user's *disclosure* in period 1.

Payoffs. The information is valuable for the platform that holds it and costly for the user, but only on the platform where she makes transactions. When the user uses platform i in period t, her payoff is

$$v(a_t) - \lambda\left(\sigma_0^2 - \sigma_{i,t}^2\right) \;=\; v(a_t) - c\, y_{i,t}, \qquad c := \lambda\sigma_0^2,$$

and platform i gets $\sigma_0^2 - \sigma_{i,t}^2 = \sigma_0^2\, y_{i,t}$. A platform that the user does not use gets nothing. The parameter c is the user's cost of full disclosure. Payoffs are measured relative to a transaction where no information is used, so they can be negative. The user always uses some platform, because the transaction itself is valuable for her (in the micro-foundation in Appendix A, her surplus is still positive under full disclosure). If the user switches from A to E in period 2, she pays a switching cost $s \in [0, \infty]$. The discount factors are $\beta_u \in (0,1)$ for the user and $\beta_p \in (0,1)$ for the platforms.

Appendix A gives a simple micro-foundation for these payoffs. In each period the user buys a product on the platform she uses, and the sellers on that platform use the platform's information to price discriminate. With linear demand, the seller's expected profit goes up by $\frac{1}{4}(\sigma_0^2 - \sigma^2)$ and the buyer's expected surplus goes down by $\frac{3}{8}(\sigma_0^2 - \sigma^2)$, where σ^2 is the posterior variance of the platform where the transaction happens. The platform gets the seller's gain. Two features are important here. The information is *persistent*, because the platform uses its stock in all later transactions. And the information is *transactional*, because the data of a platform that the user has left does not affect her anymore, while a new platform starts with no data.

3.2 Timing and equilibrium

1. *Period 1.* A announces γ_1; the user chooses a_1; A observes z_1; payoffs are realized.
2. *Period 2.* A and E announce $\gamma_{A,2}$ and $\gamma_{E,2}$ at the same time; the user chooses a platform (and pays s if she switches) and an activity level a_2; the chosen platform observes z_2; payoffs are realized.

All actions are observed by everyone. A platform can only commit to its protection level for one period, so in period 1 the incumbent cannot commit to its protection in period 2 (this is the one-period commitment case in Ichihashi, 2023). Monopoly corresponds to $s = \infty$, and a fully contestable market corresponds to $s = 0$.

The signals affect payoffs only through the posterior variances, and the posterior variances only depend on the actions. So the game is basically a game of complete information, and we use subgame perfect equilibrium (SPE) with two standard selection rules. (E1) When the user is indifferent, she chooses what the platforms prefer. Between activity levels she chooses the one preferred by the platform she uses, and between platforms she chooses the one that can offer her a higher maximal payoff (the incumbent, if both can offer the same). This is the usual tie-breaking rule in Bertrand games with asymmetric costs. (E2) If several protection levels give a platform the same payoff, it chooses the one that gives the highest activity. This is the limit of the case where platforms also get a very small revenue from engagement. Under these rules the equilibrium outcome is unique, so we just call it *the* equilibrium.

3.3 Assumptions

Let $\phi(a) := a\sqrt{v'(a)}$.

Assumption 1 (Enjoyment)**.** *v is twice continuously differentiable on $[0, M]$, $v(0) = 0$, $v' > 0$ on $[0, M)$, $v'(M) = 0$, $v'' < 0$, and v' is log-concave.*

Enjoyment is concave and the user gets satiated at M, so without privacy concerns she would choose M. The log-concavity of v' (for example, $v''' \leq 0$ is enough) makes sure that the function $a \mapsto a^2 v'(a)$ has a unique maximizer $\tilde{a} \in (0, M)$, so ϕ is increasing on $[0, \tilde{a}]$ and decreasing on $[\tilde{a}, M]$ (Lemma B.1). Let $\phi^* := \phi(\tilde{a})$ and define

$$\tilde{x} := \frac{\phi^*}{\sqrt{\lambda}}, \qquad \kappa := 1 - \tilde{x}, \qquad \Delta := v(M) - v(\tilde{a}) > 0.$$

The quadratic function $v(a) = \frac{K}{2M} a(2M - a)$ satisfies Assumption 1, with $\tilde{a} = \frac{2}{3}M$ and $\phi^* = \frac{2}{3}M\sqrt{K/3}$.

Assumption 2 (Regularity)**.** (a) $\lambda > (\phi^*)^2$; (b) $\sigma_0^2\, \tilde{a}\, (1 - \tilde{x})^2 \geq \tilde{x}$; (c) $v(\tilde{a}) \geq \sqrt{1 + \beta_u}\; c\, \tilde{x}$.

Part (a) means that the user cares enough about privacy, so that a platform cannot learn everything in one period ($\tilde{x} < 1$). Part (b) means that the prior uncertainty is large compared with what one period of activity reveals; it makes sure that all protection levels in equilibrium are interior. Part (c) means that the platform is attractive enough, so the user always wants to be active. All three parts hold in the numerical example that we use for the figures: quadratic v with $M = 1$ and $K = 0.1$, $\lambda = 1$, $\sigma_0^2 = 0.25$, and $\beta_u = \beta_p = 0.9$, where $\tilde{x} \approx 0.122$.

4 Benchmarks

4.1 The privacy Laffer curve

Consider one period where a platform with prior variance σ^2 serves the user, and the user's cost of revealed information is Λ per unit of variance. (In a one-period problem $\Lambda = \lambda$. Later we will see that a forward-looking user acts as if $\Lambda > \lambda$.) Given protection $\gamma < \infty$, the user solves

$$S_\Lambda(\gamma; \sigma^2) \;:=\; \max_{a \in [0,M]} \; v(a) - \Lambda\, \sigma^2\, \xi(a, \gamma; \sigma^2), \tag{3}$$

where, with a small abuse of notation, $\xi(a, \gamma; \sigma^2) = a\sigma^2/(1 + a(\sigma^2 + \gamma))$. Since $\partial(\sigma^2 \xi)/\partial a = (\xi/a)^2$, an interior optimum satisfies $v'(a) = \Lambda(\xi/a)^2$, that is,

$$\xi \;=\; \frac{a\sqrt{v'(a)}}{\sqrt{\Lambda}} \;=\; \frac{\phi(a)}{\sqrt{\Lambda}}. \tag{4}$$

Note that condition (4) gives the share of uncertainty that the user reveals as a function of her activity only. It does not depend on γ or σ^2, so protection affects disclosure only through the activity level that it induces.

Lemma 1 (The user's response). *Fix $\sigma^2 > 0$, $\gamma \in [0, \infty)$, and $\Lambda > 0$. Every solution of* (3) *is either $a = 0$ or $a_\Lambda(\gamma)$, the largest solution of* (4) *(when there is one). The activity $a_\Lambda(\gamma)$ is strictly increasing in γ and strictly decreasing in Λ. The user's value $S_\Lambda(\gamma; \sigma^2)$ is nondecreasing in γ, and $\partial S_\Lambda / \partial \gamma = \phi(a_\Lambda(\gamma))^2$ when $a_\Lambda(\gamma)$ is the unique optimum. Also, any $a^\circ \in [\tilde{a}, M)$ is induced by the protection*

$$\Gamma_\Lambda(a^\circ; \sigma^2) \;:=\; \sigma^2 \frac{1 - \xi^\circ}{\xi^\circ} - \frac{1}{a^\circ}, \qquad \xi^\circ := \frac{\phi(a^\circ)}{\sqrt{\Lambda}}, \tag{5}$$

in the sense that $a_\Lambda(\Gamma_\Lambda(a^\circ; \sigma^2)) = a^\circ$, if $\Gamma_\Lambda(a^\circ; \sigma^2) \geq 0$.

The platform chooses γ to maximize the information $\sigma^2 \xi$ that it collects. By (4), any interior response of the user reveals the share $\phi(a)/\sqrt{\Lambda} \leq \phi^*/\sqrt{\Lambda}$, and the equality holds only when the activity is $\tilde{a}$. This gives the following result.

Proposition 1 (Privacy Laffer curve). *Let $\sigma^2 \in [(1 - \tilde{x})\sigma_0^2, \sigma_0^2]$ and $\Lambda \in [\lambda, (1 + \beta_u)\lambda]$, and let $\hat{x} := \phi^*/\sqrt{\Lambda}$.*

(a) The information collected by the platform is single-peaked in γ, and the user's payoff is nondecreasing in γ.

(b) The unique optimal protection of the platform is

$$\hat{\gamma}(\sigma^2, \Lambda) \;=\; \sigma^2 \frac{1 - \hat{x}}{\hat{x}} - \frac{1}{\tilde{a}} \;\geq 0. \tag{6}$$

It induces activity $\tilde{a}$, and the platform learns the share $\hat{x}$ of the user's remaining uncertainty, which does not depend on σ^2.

(c) $\hat{\gamma}$ is strictly increasing in σ^2 and in Λ.

Protection has two effects on what the platform learns. The *direct effect* is negative, because each unit of activity becomes less informative. The *engagement effect* is positive, because a user with more protection is more active. When protection is low, the engagement effect is stronger, so both the user and the platform gain from more protection, and the platform never chooses such a low protection. This is why even a monopoly platform protects privacy. Figure 1 shows an example.

The static monopoly benchmark is $\sigma^2 = \sigma_0^2$ and $\Lambda = \lambda$: the platform offers $\tilde{\gamma} := \hat{\gamma}(\sigma_0^2, \lambda)$ and learns the share $\tilde{x}$. Part (b) contains a scale invariance property, which makes the dynamic analysis tractable. The platform always learns the same share of what it does not know yet, and it adjusts its protection level to the size of the remaining uncertainty. By part (c), a platform that already knows more offers less protection.

4.2 Dynamic monopoly

Now suppose there is no rival ($s = \infty$). In period 2 the platform with stock x_1 faces the static problem with prior variance $(1 - x_1)\sigma_0^2$ and $\Lambda = \lambda$. By Proposition 1, it learns

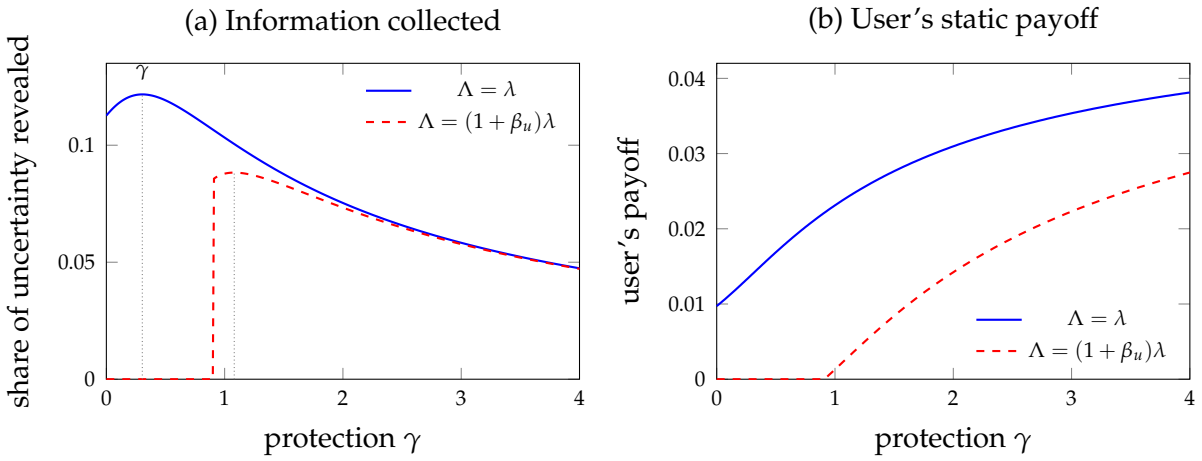


Figure 1: The privacy Laffer curve. Panel (a): the share of the user's uncertainty that the platform learns, as a function of the protection level, for an unguarded user ($\Lambda = \lambda$) and for a user with the highest caution ($\Lambda = (1+\beta_u)\lambda$). The information is highest at $\hat{\gamma}(\sigma_0^2, \Lambda)$, where the user's activity is $\tilde{a}$. The more careful user is not active at all unless protection is high enough. Panel (b): the user's payoff increases with protection. Parameters: quadratic v with $M = 1$, $K = 0.1$; $\lambda = 1$, $\sigma_0^2 = 0.25$, $\beta_u = 0.9$.

the share $\tilde{x}$ of what is left, so its stock becomes $x_1 + (1 - x_1)\tilde{x} = \tilde{x} + \kappa x_1$, and the user's payoff in period 2 is

$$u^M(x_1) \;:=\; v(\tilde{a}) - c\,(\tilde{x} + \kappa x_1). \tag{7}$$

So each unit of information disclosed in period 1 costs the user $\kappa = 1 - \tilde{x} < 1$ units in period 2, not one unit. The reason is that if the user discloses more today, there is less left for the monopoly to learn tomorrow, so part of today's data just replaces tomorrow's data. Using (7), the user's problem in period 1 is the static problem with a higher cost $\Lambda_M := \lambda(1 + \beta_u\kappa)$. The platform's objective $\sigma_0^2[x_1 + \beta_p(\tilde{x} + \kappa x_1)]$ is increasing in x_1, so the platform just maximizes the data it collects in period 1.

Proposition 2 (Dynamic monopoly)**.** *If $s = \infty$, the equilibrium is unique. The user's activity is $\tilde{a}$ in both periods. In period 1 the platform learns the share*

$$x^M \;:=\; \frac{\tilde{x}}{\sqrt{1 + \beta_u\kappa}} \;<\; \tilde{x}$$

of the prior uncertainty, and in period 2 it learns the share $\tilde{x}$ of the remaining uncertainty. The protection levels satisfy

$$\gamma_1 = \hat{\gamma}(\sigma_0^2, \Lambda_M) \;>\; \tilde{\gamma} \;>\; \gamma_2 = \hat{\gamma}\big((1 - x^M)\sigma_0^2, \lambda\big),$$

and none of these results depends on the platform's discount factor β_p.

So privacy protection is higher at the beginning and then goes down, which is similar to the equilibria with decreasing protection in Ichihashi (2023). This result is

important for the rest of the paper. The monopoly platform is in fact myopic. Today's data leaves less to learn tomorrow, but more information today always means a larger stock tomorrow ($\tilde{x} + \kappa x_1$ is increasing in x_1), so the platform's discount factor does not matter. Protection is high in period 1 only because the user is careful. She knows that her data from period 1 will be used against her in period 2, so she acts as if privacy were more costly. The platform responds with more protection and keeps her activity at $\tilde{a}$.

We measure how careful the user is by her *exposure*, which is the rate at which one unit of disclosure in period 1 lowers her payoff in period 2, divided by c. Under monopoly the exposure is κ. In the next section we show that competition can change the exposure in both directions. Competition also gives the incumbent a reason to limit its data collection, which a monopoly platform never has.

5 Competition

5.1 Period 2: data as a liability

Fix the incumbent's stock x_1. In period 2 each platform offers a protection level, and each offer gives a payoff to the user and a stock of information to the platform. The most generous offer of the incumbent is full protection, $\gamma_{A,2} = \infty$. Then the user enjoys $v(M)$, but she still bears the cost $c\,x_1$ of the data that the incumbent already has. The most generous offer of the rival is also full protection, which is worth $v(M) - s$ to the user after she pays the switching cost. So

$$\text{incumbent's best offer: } v(M) - c\,x_1, \qquad \text{rival's best offer: } v(M) - s. \tag{8}$$

This means that the incumbent's information is a *liability* in this competition: the more it knows, the worse is the best deal it can offer. As in a Bertrand game with asymmetric costs, the platform with the better best offer serves the user. It has to match the best offer of the other platform, unless its own monopoly offer is already better for the user.

Lemma 2 (Period-2 competition)**.** *Fix $x_1 \in [0, \tilde{x}]$ and $s \in [0, \infty]$. In the period-2 subgame:*

(a) The incumbent serves the user if and only if $c\,x_1 \leq s$.

(b) If the incumbent serves, the user gets $\max\{u^M(x_1), v(M) - s\}$. The incumbent's stock is $\tilde{x} + \kappa x_1$ if $u^M(x_1) \geq v(M) - s$, and otherwise it is $y_A(x_1, s)$, the largest stock such that the incumbent can still offer the user $v(M) - s$. The function y_A is continuous and increasing in both arguments, and $y_A(s/c, s) = s/c$.

(c) If the rival serves, the user gets $v(M) - c\,x_1$, and the rival's stock is $y_E(x_1, s) < x_1$, which is decreasing in s.

In particular, the user's continuation payoff is

$$W(x_1; s) \;=\; \max\Big\{\; v(\tilde{a}) - c\,(\tilde{x} + \kappa x_1)\;,\;\; v(M) - \max\{s,\; c\,x_1\}\;\Big\}. \tag{9}$$

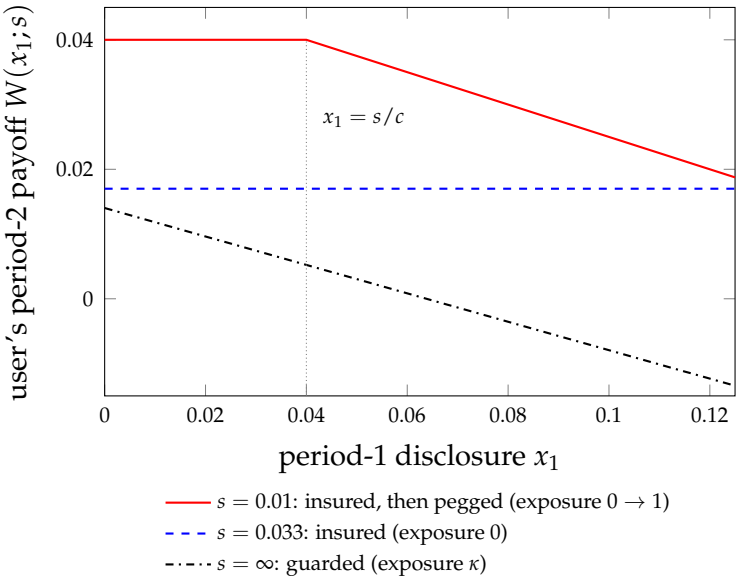


Figure 2: The user's continuation payoff (9) for three switching costs. With $s = 0.01$, the user is insured by the rival's offer as long as the incumbent's data liability $c\,x_1$ is below s, and her payoff is pegged to her disclosure when it is above s. With $s = 0.033$, she is insured for all relevant disclosures. Under monopoly she is guarded and the slope is $-c\kappa$. Same parameters as Figure 1, $c = 0.25$.

Equation (9) shows how the disclosure in period 1 affects the outcome in period 2. The three linear pieces correspond to three cases (see Figure 2):

- *Guarded* ($u^M(x_1) \geq v(M) - s$): the switching cost is so high that the rival does not constrain the incumbent, and the incumbent acts like a monopoly. The exposure is κ.
- *Insured* ($u^M(x_1) < v(M) - s$ and $c\,x_1 \leq s$): the incumbent keeps the user, but it has to match the rival's best offer $v(M) - s$, which does not depend on the user's history. The exposure is 0.
- *Pegged* ($c\,x_1 > s$): the incumbent's data liability is larger than the switching cost, so the rival takes the user. The rival collects as much information as it can while still beating the incumbent's best offer, so the user gets exactly $v(M) - c\,x_1$. Her payoff depends on what she disclosed to the incumbent, even though she leaves it. The exposure is 1.

The pegged case is the most interesting one. By switching, the user escapes the incumbent's data, but not the effect of having disclosed it: the data determines her outside option, and the rival gets all the value of her fresh start. So her exposure is larger than under monopoly, where part of today's disclosure only replaces tomorrow's.

5.2 Exposure

We define the user's exposure at disclosure x_1 as $e(x_1; s) := -\frac{1}{c}\,\partial W(x_1; s)/\partial x_1 \in \{\kappa, 0, 1\}$. When the exposure is constant around the equilibrium, the user's problem in period 1 is the static problem with cost $\lambda(1 + \beta_u e)$, and we can find her disclosure in closed form.

Lemma 3 (Exposure)**.** *Suppose that in equilibrium the user's disclosure in period 1, x_1^*, is in the interior of an interval where $e(\cdot; s)$ is constant and equal to e, that her best response to the equilibrium protection level is unique, and that the incumbent maximizes the disclosure in period 1. Then $a_1 = \tilde{a}$,*

$$x_1^* \;=\; \frac{\tilde{x}}{\sqrt{1+\beta_u e}}, \qquad \textit{and} \qquad \gamma_1 = \hat{\gamma}\big(\sigma_0^2, \lambda(1+\beta_u e)\big). \tag{10}$$

Exposure $e = 1$ gives $x^P := \tilde{x}/\sqrt{1+\beta_u}$, exposure κ gives the monopoly level x^M, and exposure 0 gives the static level $\tilde{x}$, so $x^P < x^M < \tilde{x}$. But the analysis is more than a comparison of exposures, for three reasons. First, the incumbent does not always want to maximize disclosure: it loses the user when $c\,x_1 > s$, so it may prefer to collect less. Second, the user's disclosure can be at a kink of W, where the exposure jumps. Third, even when W is flat, the user may be indifferent between two pieces of W, for example between being active as an insured user and becoming guarded. Then the incumbent has to change its protection to keep her on the piece it prefers. The main result takes all three into account.

5.3 Main result

We use the following thresholds on the switching cost:

$$s_P := \frac{c\,x^P}{1+\beta_p}, \qquad s_1 := c\,\tilde{x}, \qquad s_2 := \Delta + c\,\tilde{x}, \qquad s_3 := \Delta + (1+\kappa)\,c\,\tilde{x}.$$

It is easy to see that $0 < s_P < s_1 < s_2 < s_3$. When $s \in (s_2, s_3)$, both the guarded piece and the insured piece of W matter, and in period 1 the user chooses between two static problems: an *insured* problem with cost λ and continuation $v(M) - s$, and a *guarded* problem with cost Λ_M and continuation u^M. The value of being insured at protection γ is the difference between the two static values,

$$R(\gamma) \;:=\; S_\lambda(\gamma; \sigma_0^2) - S_{\Lambda_M}(\gamma; \sigma_0^2) \;\geq 0,$$

and we define

$$\bar{s} := s_2 + \frac{R(\tilde{\gamma})}{\beta_u}, \qquad \hat{s} := s_2 + \frac{\max_{\gamma \geq \tilde{\gamma}} R(\gamma)}{\beta_u},$$

and, for $s \in (\bar{s}, \hat{s}]$, $\gamma^*(s) := \min\{\gamma \geq \tilde{\gamma} :\; R(\gamma) \geq \beta_u(s - s_2)\}$. Finally, let $x_\lambda(\gamma) := \xi(a_\lambda(\gamma), \gamma; 0)$ be the disclosure of an unguarded user who faces protection γ.

Theorem 1 (Competition and privacy)**.** *Under Assumptions 1 and 2, $s_2 < \bar{s} < \hat{s} < s_3$, and the equilibrium is as follows.*

(i) ***Poaching*** *($s < s_P$): the incumbent induces $a_1 = \tilde{a}$ and $x_1^* = x^P$. The rival serves the user in period 2, and she gets $v(M) - c\,x^P$.*

(ii) ***Limit collection*** *($s_P \leq s \leq s_1$): the incumbent induces $x_1^* = s/c$, with activity $a_1 = \phi^{-1}\big(\sqrt{\lambda}\,s/c\big) \geq \tilde{a}$ on the decreasing part of ϕ. It keeps the user in period 2 without collecting any new data, and she gets $v(M) - s$.*

(iii) ***Insured incumbency*** *($s_1 \leq s \leq \bar{s}$): the incumbent offers the static protection $\tilde{\gamma}$ and $x_1^* = \tilde{x}$. It keeps the user in period 2, and she gets $v(M) - s$.*

(iv) ***Transition*** *($\bar{s} < s \leq \hat{s}$): the incumbent offers $\gamma^*(s) > \tilde{\gamma}$ and $x_1^* = x_\lambda(\gamma^*(s)) \in (x^M, \tilde{x})$, which is continuous and strictly decreasing in s. It keeps the user in period 2, and she gets $v(M) - s$.*

(v) ***Monopoly*** *($s > \hat{s}$): the outcome is the same as in Proposition 2, with $x_1^* = x^M$.*

As a result, the disclosure in period 1 is non-monotone in s. It jumps down at s_P and just above $\hat{s}$, it is continuous everywhere else, and

$$\min_s x_1^*(s) \;=\; \frac{x^P}{1+\beta_p} \;<\; x_1^*(0) = x^P \;<\; x_1^*(\infty) = x^M \;<\; \max_s x_1^*(s) = \tilde{x}.$$

The protection γ_1 in period 1 moves in the opposite direction of x_1^*.

Table 1 and Figure 3 summarize the theorem. The five regimes combine the user side and the platform side as follows.

Poaching. When switching is cheap, the incumbent can keep the user only if its data liability $c\,x_1$ stays below s, and this means collecting very little data. The incumbent prefers to collect as much as possible in period 1 and lose the user. The user knows that the rival's terms will depend on what she discloses, so her exposure is one and she is more careful than under monopoly. So contestability increases privacy in period 1, even though the incumbent knows that it will lose the user and has no interest in the future relationship. One may think that an incumbent that expects to lose the user should lower its protection, but this ignores the reaction of the user.

Limit collection. For intermediate switching costs, the incumbent can keep the user if it limits its data collection in period 1 to $x_1 = s/c$, because then its best offer just matches the rival's offer. It does so when keeping the user is worth more than the extra data it could collect by letting her go, that is, when $(1+\beta_p)s/c \geq x^P$, or $s \geq s_P$. This is a retention motive on the platform side, which a monopoly never has, and it depends on the platform's patience. In this regime privacy goes *down* when switching becomes more costly, because the incumbent's data limit s/c goes up. The user's disclosure is at the kink of W: the incumbent chooses protection so that an insured user would disclose exactly s/c, and the risk of being pegged stops her from disclosing more. The incumbent then keeps her in period 2 with full protection and does not collect any new

Table 1: Equilibrium regimes (Theorem 1)

Regime	Switching cost	Disclosure x_1^*	Exposure	Period 2
Poaching	$s < s_P$	$\tilde{x}/\sqrt{1+\beta_u}$	1 (pegged)	rival serves; $W = v(M) - c\,x_1^*$
Limit collection	$s_P \le s \le c\,\tilde{x}$	s/c	kink	incumbent serves, no new data; $W = v(M) - s$
Insured incumbency	$c\,\tilde{x} \le s \le \bar{s}$	$\tilde{x}$	0 (insured)	incumbent serves; $W = v(M) - s$
Transition	$\bar{s} < s \le \hat{s}$	$x_\lambda(\gamma^*(s))$, decreasing	0 (insured)	incumbent serves; $W = v(M) - s$
Monopoly	$s > \hat{s}$	$\tilde{x}/\sqrt{1+\beta_u\kappa}$	κ (guarded)	incumbent serves, unconstrained; $W = u^M(x_1^*)$

data. By rule (E2), the incumbent implements the limit with the highest protection, which gives the highest engagement.

Insured incumbency. When $s \ge s_1 = c\,\tilde{x}$, the limit does not bind anymore: even with the static disclosure $\tilde{x}$, the incumbent's best offer is above the rival's. The user's payoff in period 2 is fixed at $v(M) - s$ by the rival's offer, and it does not depend on her history. So her exposure is zero and she discloses the static amount. In this case the incumbent offers exactly the static monopoly protection $\tilde{\gamma}$. The reason is not that it has no interest in the future, but that the user is insured.

Transition and monopoly. When s gets larger, being insured is less valuable. The user compares being active as an insured user (with continuation $v(M) - s$) with being active as a guarded user under an unconstrained incumbent. She prefers the second option when $\beta_u(s - s_2)$ is larger than the value of being insured, $R(\gamma)$. To keep her insured, the incumbent increases protection, which lowers her disclosure. When s is larger than $\hat{s}$ this is not possible anymore, and we get the monopoly outcome. In our numerical example, in almost all of the transition regime the guarded option that binds is not being active at all ($a_1 = 0$), so the user is indifferent between being active as an insured user and not being active. Only in a very small interval at the top of the regime, the binding option is to be active as a guarded user.

5.4 Discussion

Theorem 1 shows that "more competition" is not enough to know what happens to privacy. What matters is how competition changes two things: the user's exposure, which determines how careful she is with her data, and the incumbent's interest in keeping her, which determines how much it limits itself.

The comparison between monopoly and full contestability ($s = \infty$ versus $s = 0$) is subtle. In both cases the incumbent maximizes data collection in period 1: the monopoly because it is myopic, and the contested incumbent because it expects to lose the user anyway. So the difference only comes from the user's exposure, κ versus 1. This difference is small when $\tilde{x}$ is small, because the monopoly only learns a small share of

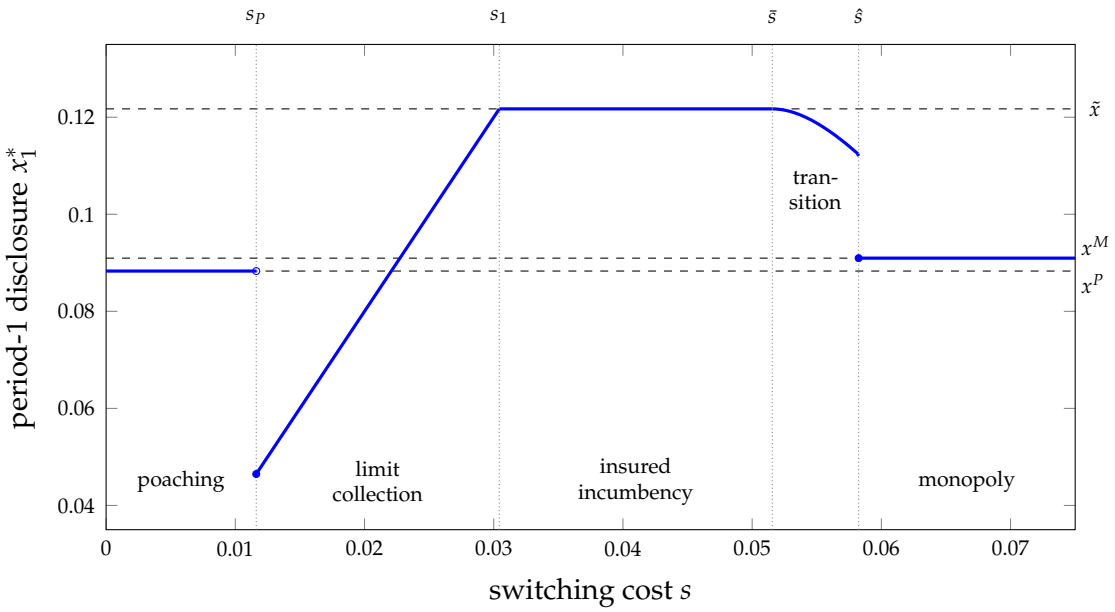


Figure 3: Equilibrium disclosure in period 1 as a function of the switching cost (Theorem 1). Lower disclosure means more privacy. The dashed lines show the disclosure levels $\tilde{x}/\sqrt{1+\beta_u e}$ for exposure $e = 1$ (x^P), $e = \kappa$ (x^M) and $e = 0$ ($\tilde{x}$). Same parameters as Figure 1 and $\beta_p = 0.9$. The thresholds are $s_P \approx 0.0116$, $s_1 \approx 0.0304$, $\bar{s} \approx 0.0516$, and $\hat{s} \approx 0.0582$.

what is left in each period. The large effects of competition come from the other two channels: the insurance from moderate switching costs (which lowers privacy) and the incumbent's retention motive (which raises privacy).

It is also useful to compare our results with the ratchet effect in labor markets. There, competition from other employers removes the ratchet effect, because the future wage of a worker is decided by the market and not by the information of the current employer (Kanemoto and MacLeod, 1992). In our insured regime the logic is the same, and the user is not careful anymore. But in the pegged regime the rival's offer depends on the incumbent's information, and competition makes the user more careful. The difference comes from the fact that the incumbent's information is a liability: it determines the user's outside option, and not her value to the market.

6 Welfare and Policy

6.1 Who gains from competition?

Let $U(s)$ be the user's lifetime payoff in equilibrium and $\Pi_A(s)$ the incumbent's equilibrium profit, as functions of the switching cost.

Proposition 3 (Welfare). *(a) Under rule (E2), the user's payoff U is highest at $s = s_P$. It*

is constant on $[0, s_P)$, jumps up at s_P, and is strictly decreasing on $[s_P, \bar{s}]$. In particular, $U(s_P) > U(0)$ and $U(s_P) > U(\infty)$.

(b) For all $s \in [\bar{s}, \hat{s}]$, $U(s) < U(\infty)$, so for high but finite switching costs the user is worse off than under monopoly.

(c) The incumbent's profit Π_A is nondecreasing on $[0, \bar{s}]$, and $\Pi_A(\hat{s}) > \Pi_A(\infty)$, so the incumbent prefers some competition to a full lock-in.

Figure 4 shows these results. Part (a) says that the user is best off with a small but *positive* switching cost, which is the smallest one at which the incumbent prefers to keep her rather than to let her go. At this point three things happen together: the incumbent collects much less data in period 1 than in the poaching case, it gives the user more protection (so she is more active), and it serves her in period 2 without collecting new data. Without switching costs, instead, the incumbent collects as much data as it can, and the user goes to a rival whose offer depends on this data. Note that this comparison uses rule (E2). In the limit-collection regime the incumbent is indifferent between all protection levels that give the data limit s/c, and (E2) picks the most protective one, which gives the highest engagement. This is natural when platforms get even a small revenue from engagement, like ad-funded platforms. With the least protective choice, the user's payoff just above s_P can be lower than at $s = 0$, but it is still much higher than under monopoly. (In the numerical example it goes from about 0.073 to about 0.041, while it is 0.048 at $s = 0$.)

Part (b) is more surprising. For switching costs between $\bar{s}$ and $\hat{s}$ (and also a bit below $\bar{s}$), the user is worse off than under monopoly, even though she is served by the same incumbent and she has a better outside option. The reason is that a weak outside option insures her. Her payoff in period 2 does not depend on what she discloses, so she stops being careful and discloses more in period 1, while the insurance itself is not worth much. In the transition regime, the incumbent chooses its protection exactly to keep her indifferent to becoming guarded, so she gets none of the value of the insurance.

Part (c) is the other side of the same story, and it is similar to the idea of second sourcing (Shepard, 1987; Farrell and Gallini, 1988). A credible rival commits the incumbent to leave the user at least $v(M) - s$ in period 2. Since the user is insured, she is less careful and discloses more in period 1, and the incumbent gains more from this than it loses from being constrained later. So if a platform could choose its switching costs, for example by supporting interoperability or data portability, it would not choose a full lock-in. In the numerical example, the incumbent's profit at $s = \hat{s}$ is about 13% higher than under monopoly.

6.2 Policy implications

Lowering switching costs. Interoperability rules and data portability lower the switching cost. Theorem 1 and Proposition 3 show that the effect of these policies on privacy and on users depends on how much the switching cost goes down. Starting from lock-in, a moderate reduction can *reduce* privacy and hurt users. The reason is that

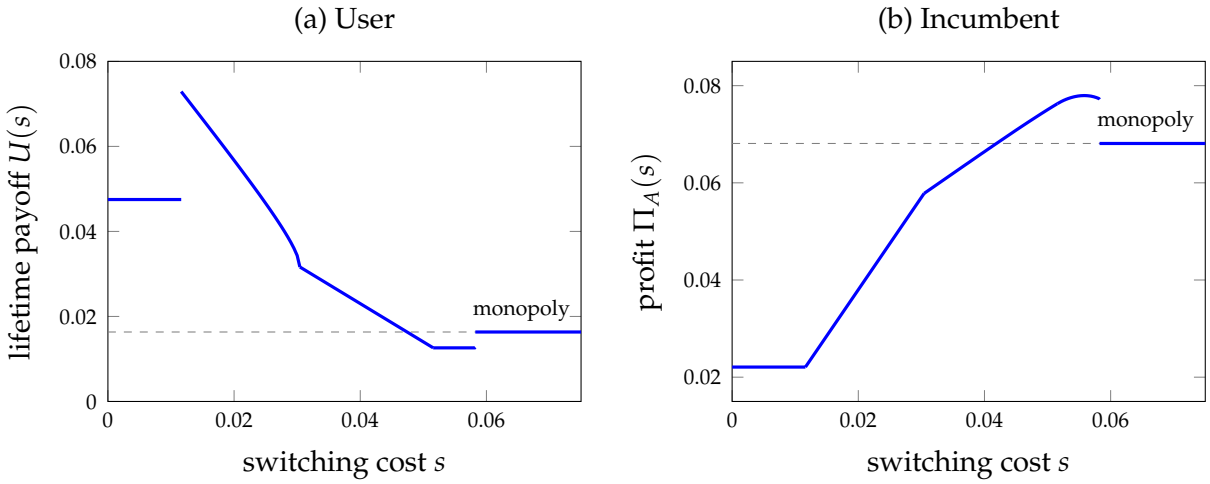


Figure 4: Welfare as a function of the switching cost (Proposition 3). The user is best off at $s = s_P$, where the incumbent starts to limit its data collection, and she is worse off than under monopoly for high but finite switching costs. The incumbent's profit is highest in the transition regime, where it is higher than its monopoly profit. Same parameters as Figure 3.

the market moves to the transition or insured regimes, where users are insured by a weak outside option and stop protecting their data. A larger reduction, which brings the switching cost below the incumbent's data liability $c\,\tilde{x}$, makes the incumbent afraid of losing the user and leads to limit collection. Then the user is better off than under lock-in, and privacy is higher than under lock-in once $s < c\,x^M$. But going all the way to zero switching costs is too much: the incumbent gives up on keeping the user and collects as much as it can. So we should evaluate these policies by where they bring the market compared with the incumbent's data liability, and not only by whether they make the market more contestable.

Erasure rights. Privacy laws like the GDPR give users the right to have their data erased. In the model, suppose that at the beginning of period 2 the user can ask the incumbent to erase her data at no cost, so that its stock goes back to zero.

Proposition 4 (Erasure right)**.** *With an erasure right, for every s the equilibrium has $a_1 = \tilde{a}$, $x_1^* = \tilde{x}$, and $\gamma_1 = \tilde{\gamma}$, the incumbent serves the user in period 2, and she gets* $\max\{v(\tilde{a}) - c\,\tilde{x},\ v(M) - s\}$*. Compared with the equilibrium without the right, the user is* (a) *better off under poaching ($s < s_P$) if and only if $s < s_R := c\,\tilde{x}(\sqrt{1+\beta_u} - 1)/\beta_u$;* (b) *strictly worse off under limit collection ($s_P \leq s < s_1$);* (c) *equally well off if $s \in [s_1, s_2]$; and* (d) *strictly better off if $s > s_2$.*

An erasure right removes the user's exposure, because what she discloses today can be deleted tomorrow. So she stops being careful and discloses the static amount $\tilde{x}$ in every market structure, and by this measure privacy goes *down*. But she is better

off when her caution was costly and did not give her much. This is the case under monopoly, in the transition regime, in the upper part of the insured regime ($s > s_2$), and under poaching when the switching cost is small. (In the numerical example $s_R > s_P$, so this holds in the whole poaching regime.) The right hurts her when competition was already disciplining the incumbent. The clearest case is that the incumbent has no reason to limit its data collection anymore: with erasure its data is not a liability, so it is never afraid of losing the user. If platforms are impatient enough that $s_R < s_P$, the right can also hurt the user under poaching, when escaping the pegged terms is not worth much. For the same reason, the incumbent benefits from the erasure right when switching costs are low, because the right removes the threat of poaching. There are two lessons here. First, measured privacy (how much users disclose) and user welfare can go in opposite directions. Second, data rights and competition policy interact: a right that protects users in a concentrated market can weaken the discipline that competition gives.

7 Extensions and Discussion

7.1 Competition for the market

In the baseline model, the incumbent is alone in period 1, and competition only comes after some data has been collected. Now suppose instead that two symmetric platforms without data compete for the user in period 1, and the one she chooses becomes the incumbent: in period 2 she can switch to the other platform at cost s. This is competition *for* the market, followed by lock-in.

Proposition 5 (Competition for the market)**.** *Suppose both platforms are available in period 1 and $s \geq s_1$. In every equilibrium, the platform that the user chooses in period 1 offers full protection, so $x_1^* = 0$, and it serves her again in period 2, where she gets* $\max\{v(\tilde{a}) - c\,\tilde{x},\ v(M) - s\}$. *If $s \geq s_2$, the platform in period 2 acts as an unconstrained monopoly and learns the share $\tilde{x}$ of her uncertainty.*

When platforms compete before any data exists, they compete with privacy. Each platform tries to win the user by promising more protection. Since the prize (the right to use a locked-in user tomorrow) is positive even with full protection, competition pushes data collection in period 1 to zero. Once the user is locked in ($s \geq s_2$), the winner uses her like a monopoly. This is a privacy version of the "bargains then rip-offs" pattern in markets with switching costs (Klemperer, 1987; Farrell and Klemperer, 2007). It also fits the story where a platform wins the market by promising not to track its users and then changes its policy later (Srinivasan, 2019). Compared with the baseline model, the timing of competition relative to data collection determines how privacy changes over time. Competition before lock-in makes privacy high at the beginning, while competition after data is collected affects early privacy in a non-monotone way, through exposure and limit collection.

Remark 1 (Symmetric competition in both periods)**.** If $s = 0$ and both platforms are available in both periods, the same Bertrand logic works in both periods. In period 2, a platform with data loses the user to a platform without data, and two platforms without data compete away all the information. In period 1, winning is worth more than losing for any positive disclosure, because a poacher collects less than the period-1 winner. So competition pushes data collection to zero in both periods. To get positive data collection with symmetric competition, we would need an exogenous upper bound on the protection that platforms can offer.

7.2 Robustness and limitations

Imperfect competition. With Bertrand competition the regimes are sharp. If platforms were horizontally differentiated, or if the switching cost were random from the platforms' point of view, the incumbent would keep the user with a probability that decreases in its data stock. Then the user's exposure would be a weighted average of 1 (pegged), 0 (insured) and κ (guarded), and the retention motive of the incumbent would work at the margin and not at a kink. Both forces in this paper would still be there, but limit collection would become a smooth trade-off between data today and retention tomorrow.

Where information hurts. The transactional interpretation is important for the liability effect: information hurts the user only where she makes transactions. If data hurt the user no matter which platform she used, for example because it is sold to data brokers, switching would not reset her exposure. Then the rival would not have the advantage of a fresh start, and the incumbent's data would not be a liability in the competition for the user anymore, because it would lower her payoff in the same way on both platforms.

Data sharing. A related case is a policy that lets the rival get the incumbent's data (portability as data transfer, not as lower switching costs). Then both platforms can offer at most $v(M) - c\,x_1$, the incumbent keeps the user, and her payoff is pegged to her disclosure whenever the rival constrains the incumbent. Her exposure would be one instead of zero. So data sharing makes users more careful, but it also removes the fresh start that competition otherwise gives.

Commitment and horizon. In our model platforms commit to protection only one period at a time. With long-run commitment, a monopoly platform could promise future protection and make the user less careful, as in Ichihashi (2023). In our model the rival's offer in the insured regime plays exactly this role, and this is why the incumbent may like some competition. We use two periods to keep the analysis simple. Because of the scale invariance in Proposition 1, the continuation values under monopoly are linear in the information stock. So the exposure still summarizes the user's caution with more periods, until the protection level reaches zero.

8 Conclusion

This paper studies how competition affects the privacy protection of platforms when data is persistent and platform specific. There are two main forces. On the user side, the user protects her data according to her exposure, which is how much today's disclosure will cost her tomorrow. Market structure can change the exposure in both directions: a rival whose offer depends on the incumbent's data increases it, and a rival whose offer insures the user removes it. On the platform side, competition creates a retention motive that a monopoly never has: an incumbent whose data is a liability limits its data collection in order to keep its users. Together, these forces make privacy non-monotone in switching costs, make users best off with a small but positive switching cost, and can make the incumbent prefer some competition to a full lock-in.

There are several directions for future work. With many users, data externalities between users would interact with the liability effect in this paper. Multi-homing would let users split their activity between platforms, and this changes what each platform learns. Finally, some predictions of the model could be tested with data. For example, privacy policies should become more protective when the incumbent's data liability gets close to the users' switching costs, and small reductions in switching costs can reduce privacy.

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A Micro-foundation of payoffs

In this appendix we give a simple pricing foundation for the payoffs in Section 3. In each period the user buys a product from a seller on the platform she uses. Her utility from quantity q at unit price p is $Xq - \frac{1}{2}q^2 - pq$, so her demand is $q = X - p$ (as usual in linear–quadratic Gaussian models, we ignore the constraint $q \geq 0$). The seller has zero cost and uses the platform's information to set the price. It knows the posterior mean $m := \mathbb{E}[X \mid \text{platform's signals}]$, which has mean μ and variance $V := \operatorname{Var}(m) = \sigma_0^2 - \sigma^2$, where σ^2 is the platform's posterior variance. The optimal price is $p = m/2$.

The buyer's surplus is $q^2/2$ with $q = X - m/2$. Since $\operatorname{Cov}(X, m) = \operatorname{Var}(m) = V$, we have

$$\mathbb{E}\Big[\tfrac{1}{2}q^2\Big] = \tfrac{1}{2}\Big(\operatorname{Var}(X) - \operatorname{Cov}(X, m) + \tfrac{1}{4}\operatorname{Var}(m)\Big) + \tfrac{1}{8}\mu^2 = \tfrac{1}{2}\sigma_0^2 + \tfrac{1}{8}\mu^2 - \tfrac{3}{8}\,(\sigma_0^2 - \sigma^2).$$

The seller's expected profit is

$$\mathbb{E}\Big[\tfrac{m}{2}\big(X - \tfrac{m}{2}\big)\Big] = \tfrac{1}{2}\,\mathbb{E}[mX] - \tfrac{1}{4}\,\mathbb{E}[m^2] = \tfrac{1}{4}\big(\mu^2 + \sigma_0^2 - \sigma^2\big).$$

Compared with a seller without information, the buyer loses $\frac{3}{8}(\sigma_0^2 - \sigma^2)$ and the seller gains $\frac{1}{4}(\sigma_0^2 - \sigma^2)$, so total surplus goes down by $\frac{1}{8}(\sigma_0^2 - \sigma^2)$. Even with full disclosure ($\sigma^2 = 0$), the buyer's expected surplus $\frac{1}{8}(\sigma_0^2 + \mu^2)$ is positive, so she prefers to buy on some platform rather than not to buy at all. If the platform gets the seller's gain, for example by selling targeted access, its revenue is proportional to $\sigma_0^2 - \sigma^2$. Since the seller uses the information of the platform where the transaction happens, the user is only affected by the information of the platform she is using, while the platform keeps its information for later periods. Adding the enjoyment from activity, we get the payoffs of Section 3 with $\lambda = 3/8$ (the platform's payoff is multiplied by a constant, which does not change its decisions).

B Proofs

B.1 Preliminaries

Lemma B.1. Under Assumption 1:

(a) There is a unique $\tilde{a} \in (0, M)$ such that $a \mapsto a^2 v'(a)$ is strictly increasing on $(0, \tilde{a}]$ and strictly decreasing on $[\tilde{a}, M)$. Hence ϕ is strictly increasing on $[0, \tilde{a}]$, strictly decreasing on $[\tilde{a}, M]$, and $\phi(0) = \phi(M) = 0$. Moreover, $2v'(a) + av''(a) \leq 0$ for $a \geq \tilde{a}$.

(b) For any $\sigma^2 > 0$ and $\gamma \geq 0$, the function $h(a) := \sqrt{v'(a)}\,\big(1 + a(\sigma^2 + \gamma)\big)$ is strictly log-concave on $[0, M)$, and $h(M) = 0$.

Proof. (a) On $(0, M)$, $\frac{d}{da}\log\big(a^2 v'(a)\big) = \frac{2}{a} + \frac{v''(a)}{v'(a)}$. The first term is strictly decreasing and the second is nonincreasing because v' is log-concave. The sum tends to $+\infty$ as

$a \downarrow 0$ and to $-\infty$ as $a \uparrow M$, since $v'(a) \to 0$ while $v''(M) < 0$. Hence it has a unique zero $\tilde{a}$, is positive before it, and is negative after it, which also gives $2v' + av'' \leq 0$ on $[\tilde{a}, M)$. (b) $\log h = \frac{1}{2}\log v' + \log(1 + a(\sigma^2 + \gamma))$ is the sum of a concave and a strictly concave function. □

Lemma B.2 (Technology)**.** For $\sigma^2 > 0$, $\gamma \in [0, \infty)$, and $a > 0$, write $\xi = \xi(a, \gamma; \sigma^2) = a\sigma^2/(1 + a(\sigma^2 + \gamma))$. Then

$$\frac{\xi}{a} = \frac{\sigma^2}{1 + a(\sigma^2 + \gamma)}, \qquad \frac{\partial(\sigma^2\xi)}{\partial a} = \left(\frac{\xi}{a}\right)^2, \qquad \frac{\partial(\sigma^2\xi)}{\partial \gamma} = -\xi^2.$$

Proof. Direct differentiation of $\sigma^2\xi = a\sigma^4/(1 + a(\sigma^2 + \gamma))$. □

B.2 Proof of Lemma 1

Let $V(a) := v(a) - \Lambda\sigma^2\xi(a, \gamma; \sigma^2)$. By Lemma B.2, $V'(a) = v'(a) - \Lambda\sigma^4/(1 + a(\sigma^2 + \gamma))^2$, so $V'(a)$ has the sign of $h(a) - \sigma^2\sqrt{\Lambda}$. By Lemma B.1(b), the upper contour set $\{a : h(a) \geq \sigma^2\sqrt{\Lambda}\}$ is either empty or an interval $[a_L, a_H]$ with $a_H < M$. In the first case V is strictly decreasing and $a = 0$ is the unique maximizer. In the second, V decreases on $[0, a_L]$, increases on $[a_L, a_H]$, and decreases on $[a_H, M]$, so every maximizer lies in $\{0, a_H\}$. Using $\xi/a = \sigma^2/(1 + a(\sigma^2 + \gamma))$, the equation $h(a) = \sigma^2\sqrt{\Lambda}$ is equivalent to $\sqrt{v'(a)}\, a\sigma^2/\xi = \sigma^2\sqrt{\Lambda}$, i.e., to (4). Hence a_H is the largest solution of (4); call it $a_\Lambda(\gamma)$.

Comparative statics. For $a > 0$, h is strictly increasing in γ, and $h(a) < \sigma^2\sqrt{\Lambda}$ for all $a > a_H$. If $\gamma' > \gamma$, then $h(a_H; \gamma') > \sigma^2\sqrt{\Lambda}$, so the right endpoint of the new upper contour set exceeds a_H. Similarly, if $\Lambda' > \Lambda$, then $h(a) < \sigma^2\sqrt{\Lambda'}$ for all $a \geq a_H$, so the right endpoint falls.

Value. By Lemma B.2, $\partial V/\partial\gamma = -\Lambda\,\partial(\sigma^2\xi)/\partial\gamma = \Lambda\xi^2 \geq 0$ at every a. By the envelope theorem, S_Λ is nondecreasing, and where $a_\Lambda(\gamma)$ is optimal its derivative is $\Lambda\xi^2 = \phi(a_\Lambda(\gamma))^2$, using (4).

Converse. Let $a^\circ \in [\tilde{a}, M)$, $\xi^\circ = \phi(a^\circ)/\sqrt{\Lambda}$, and $\gamma^\circ = \Gamma_\Lambda(a^\circ; \sigma^2)$. Solving $\xi(a^\circ, \gamma^\circ; \sigma^2) = \xi^\circ$ for γ° gives exactly (5), so a° solves (4). It is the largest solution because h is strictly decreasing at a°:

$$\frac{h'(a^\circ)}{h(a^\circ)} = \frac{v''(a^\circ)}{2v'(a^\circ)} + \frac{\sigma^2 + \gamma^\circ}{1 + a^\circ(\sigma^2 + \gamma^\circ)} < \frac{v''(a^\circ)}{2v'(a^\circ)} + \frac{1}{a^\circ} = \frac{2v'(a^\circ) + a^\circ v''(a^\circ)}{2a^\circ v'(a^\circ)} \leq 0$$

by Lemma B.1(a). Hence a° is to the right of the peak of h, and $a_\Lambda(\gamma^\circ) = a^\circ$. □

B.3 Proof of Proposition 1

Let $\sigma^2 \in [(1 - \tilde{x})\sigma_0^2, \sigma_0^2]$ and $\Lambda \in [\lambda, (1 + \beta_u)\lambda]$, so $\hat{x} = \phi^*/\sqrt{\Lambda} \leq \tilde{x} < 1$. Two inequalities are used repeatedly. First, for $a^\circ \geq \tilde{a}$, $\Gamma_\Lambda(a^\circ; \sigma^2) \geq \Gamma_\Lambda(\tilde{a}; \sigma^2) = \sigma^2(1 - \hat{x})/\hat{x} - 1/\tilde{a} \geq (1 - \tilde{x})^2\sigma_0^2/\tilde{x} - 1/\tilde{a} \geq 0$, by Assumption 2(b). Second, the user prefers

any $a^\circ \geq \tilde{a}$ that satisfies (4) to $a = 0$, since $v(a^\circ) - \Lambda\sigma^2\phi(a^\circ)/\sqrt{\Lambda} \geq v(\tilde{a}) - \sqrt{\Lambda}\,\sigma_0^2\phi^* \geq v(\tilde{a}) - \sqrt{1+\beta_u}\,c\,\tilde{x} \geq 0$, by Assumption 2(c).

(b) By Lemma 1, for any γ the user chooses $a = 0$ (information 0) or $a_\Lambda(\gamma)$, which yields information $\sigma^2\phi(a_\Lambda(\gamma))/\sqrt{\Lambda} \leq \sigma^2\hat{x}$. At $\hat{\gamma} = \Gamma_\Lambda(\tilde{a};\sigma^2) \geq 0$ the user chooses $\tilde{a}$ (by the two inequalities), so the bound is attained. Any optimal γ must induce $a_\Lambda(\gamma) = \tilde{a}$, and since a_Λ is strictly increasing in γ, $\hat{\gamma}$ is the unique optimum.

(a) The user's payoff is $S_\Lambda(\gamma;\sigma^2)$, nondecreasing by Lemma 1. The set of γ at which the user engages is an upper set: if $a_\Lambda(\gamma)$ is optimal at γ, then for $\gamma' > \gamma$, $V(a_\Lambda(\gamma);\gamma') \geq V(a_\Lambda(\gamma);\gamma) \geq V(0)$, so a positive activity is optimal at γ', and it must be $a_\Lambda(\gamma')$. On this set the information collected is $\sigma^2\phi(a_\Lambda(\gamma))/\sqrt{\Lambda}$, with a_Λ strictly increasing, so it increases while $a_\Lambda(\gamma) < \tilde{a}$ and decreases afterwards (Lemma B.1(a)); outside the set it is zero.

(c) $\partial\hat{\gamma}/\partial\sigma^2 = (1-\hat{x})/\hat{x} > 0$, and $\hat{x}$ is strictly decreasing in Λ while $(1-\hat{x})/\hat{x}$ is strictly decreasing in $\hat{x}$. □

B.4 Continuation payoffs and a bound on disclosure

Lemma B.3. For every s, the user's continuation payoff $W(\cdot;s)$ is nonincreasing on $[0,1]$. Consequently, for every γ_1, any best response a_1 of the user in period 1 satisfies $a_1 \leq a_\lambda(\gamma_1)$ and $x_1 = \xi(a_1,\gamma_1;0) \leq \tilde{x}$ (with $a_1 = 0$ if (4) has no solution for $\Lambda = \lambda$).

Proof. For fixed (a,γ), the incumbent's end-of-period stock $x_1 + (1-x_1)\xi(a,\gamma;x_1) = 1 - \sigma_2^2/\sigma_0^2$, with $1/\sigma_2^2 = 1/((1-x_1)\sigma_0^2) + a/(1+a\gamma)$, is increasing in x_1. Hence every offer of the incumbent is worth less to the user when x_1 is larger, while the rival's offers do not depend on x_1. By the Bertrand argument in the proof of Lemma 2 below (which does not use $x_1 \leq \tilde{x}$), the user's continuation payoff is the larger of the loser's best offer and the winner's unconstrained offer. When the incumbent serves ($c\,x_1 \leq s$), this is $\max\{U_A^{mon}(x_1), v(M) - s\}$, where $U_A^{mon}(x_1)$ is the user's payoff under the incumbent's profit-maximizing offer. It is nonincreasing in x_1: it equals $u^M(x_1)$ as long as the incumbent's optimal protection (6) is nonnegative, and beyond that point the incumbent offers $\gamma = 0$, under which the user's payoff is decreasing in x_1 by the first observation. When the rival serves, it is $\max\{v(M) - c\,x_1,\, v(\tilde{a}) - c\,\tilde{x} - s\}$. At $c\,x_1 = s$ the left expression is at least $v(M) - s$, the value of the right one. Hence W is nonincreasing.

For the second claim, write the user's period-1 objective as $V(a) = [v(a) - c\,\xi(a,\gamma_1;0)] + \beta_u W(\xi(a,\gamma_1;0))$. By Lemma 1 (with $\Lambda = \lambda$, $\sigma^2 = \sigma_0^2$), the bracket is strictly decreasing on $[a_\lambda(\gamma_1), M]$ (on $[0,M]$ if (4) has no solution), and the second term is nonincreasing in a because ξ is increasing in a. Hence no $a > a_\lambda(\gamma_1)$ is optimal, and $x_1 \leq \phi(a_\lambda(\gamma_1))/\sqrt{\lambda} \leq \tilde{x}$. □

B.5 Proof of Proposition 2

With $s = \infty$ there is no competition in period 2. For $x_1 \in [0,\tilde{x}]$ the prior variance $(1-x_1)\sigma_0^2$ lies in $[(1-\tilde{x})\sigma_0^2, \sigma_0^2]$, so Proposition 1 (with $\Lambda = \lambda$) implies that the platform

induces $\tilde{a}$ and learns the share $\tilde{x}$ of the remaining uncertainty. Its stock becomes $\tilde{x} + \kappa x_1$ and the user obtains $u^M(x_1)$. By Lemma B.3, disclosures above $\tilde{x}$ are never chosen. For $x_1 \leq \tilde{x}$ the user's period-1 objective is

$$v(a_1) - c\,x_1 + \beta_u\,u^M(x_1) = v(a_1) - \Lambda_M\,\sigma_0^2\,\xi(a_1, \gamma_1; 0) + \beta_u\big(v(\tilde{a}) - c\,\tilde{x}\big),$$

because $c(1 + \beta_u\kappa) = \Lambda_M\sigma_0^2$. This is the static problem with cost $\Lambda_M \in [\lambda, (1+\beta_u)\lambda]$. The platform's objective, $\sigma_0^2[x_1 + \beta_p(\tilde{x} + \kappa x_1)]$, is strictly increasing in x_1, so it solves the static problem. By Proposition 1, $\gamma_1 = \hat{\gamma}(\sigma_0^2, \Lambda_M)$, $a_1 = \tilde{a}$, and $x_1 = \phi^*/\sqrt{\Lambda_M} = x^M$. The ranking of protection levels follows from Proposition 1(c): $\Lambda_M > \lambda$ gives $\gamma_1 > \tilde{\gamma}$, and $(1 - x^M)\sigma_0^2 < \sigma_0^2$ gives $\gamma_2 < \tilde{\gamma}$. None of these objects involves β_p. □

B.6 Proof of Lemma 2

Offer frontiers. Consider a platform with stock $y \in [0, \tilde{x}]$, so its prior variance lies in $[(1-\tilde{x})\sigma_0^2, \sigma_0^2]$. By Lemma 1 and the two inequalities in the proof of Proposition 1, every $a \in [\tilde{a}, M)$ can be induced by a nonnegative protection level, at which the user engages. Such an offer yields the user the payoff and the platform the stock

$$U_y(a) := v(a) - c\Big[y + (1-y)\frac{\phi(a)}{\sqrt{\lambda}}\Big], \qquad Y_y(a) := y + (1-y)\frac{\phi(a)}{\sqrt{\lambda}},$$

and full protection yields $(U_y(M), Y_y(M)) = (v(M) - c\,y,\ y)$. On $[\tilde{a}, M]$, U_y is strictly increasing and Y_y is strictly decreasing. Every other offer is dominated. An offer that induces $a \in (0, \tilde{a})$ yields the stock $Y_y(a')$, where $a' \in (\tilde{a}, M)$ satisfies $\phi(a') = \phi(a)$, but a lower user payoff than $U_y(a')$, because $v(a) < v(a')$. An offer that induces $a = 0$ yields $(-c\,y,\ y)$, dominated by full protection. Hence the frontier of feasible (payoff, stock) pairs is $\{(U_y(a), Y_y(a)) : a \in [\tilde{a}, M]\}$. Let $F_y(u) := Y_y(U_y^{-1}(u))$ for $u \in [U_y(\tilde{a}), v(M) - c\,y]$ denote the largest stock compatible with giving the user u; it is continuous and strictly decreasing, with $F_y(v(M) - c\,y) = y$. The platform's unconstrained optimum is the top of the frontier, $a = \tilde{a}$ (Proposition 1): for the incumbent it gives $(u^M(x_1), \tilde{x} + \kappa x_1)$ and for the rival $(v(\tilde{a}) - c\,\tilde{x},\ \tilde{x})$.

Bertrand competition. The incumbent's maximal offer is $\bar{u}_A = v(M) - c\,x_1$ and the rival's maximal net offer is $\bar{u}_E = v(M) - s$. Suppose $\bar{u}_A \geq \bar{u}_E$. In any equilibrium the user obtains at least $\bar{u}_E$. Otherwise, the rival could offer a net payoff slightly above hers but below $\bar{u}_E$, win her, and earn a positive stock, since $F_0(u + s) > 0$ for $u + s < v(M)$. Conversely, if she obtained more than $u^* := \max\{u^M(x_1), \bar{u}_E\}$ from the incumbent, the incumbent could lower her payoff toward u^*, still win, and collect more. If the rival served her at a net payoff $u \leq \bar{u}_E \leq \bar{u}_A$, the incumbent could win by offering slightly more, and it earns at least $\sigma_0^2 x_1 \geq 0$ from serving, strictly more than zero unless $x_1 = 0$, in which case it earns $\sigma_0^2 F_0(u') > 0$ for any $u' < v(M)$. With tie-breaking rule (E1), there is thus an equilibrium in which the rival offers full protection and the incumbent serves the user at u^*, and every equilibrium has this outcome. The incumbent's stock is $F_{x_1}(u^*)$, which equals $\tilde{x} + \kappa x_1$ if $u^* = u^M(x_1)$ and

$y_A(x_1, s) := F_{x_1}(v(M) - s)$ otherwise. If instead $\bar{u}_A < \bar{u}_E$, the same argument with the roles reversed shows that the rival serves and the user obtains $\max\{\bar{u}_A,\, v(\tilde{a}) - c\,\tilde{x} - s\}$. For $x_1 \le \tilde{x}$ we have $c(x_1 - \tilde{x}) \le 0 < \Delta + s$, so this equals $\bar{u}_A = v(M) - c\,x_1$. This proves (a) and the payoff claims in (b) and (c).

Properties of y_A and y_E. Continuity follows from the continuity of F. Since F_{x_1} is decreasing, y_A is increasing in s. For fixed a, $\partial U_{x_1}(a)/\partial x_1 = -c(1 - \phi(a)/\sqrt{\lambda}) < 0$, so the activity a that solves $U_{x_1}(a) = v(M) - s$ increases with x_1. The corresponding stock is $Y = (v(a) - v(M) + s)/c$, which is increasing in a, so y_A is increasing in x_1. At $x_1 = s/c$, $v(M) - s = \bar{u}_A$, so $y_A(s/c, s) = F_{s/c}(\bar{u}_A) = s/c$. The rival's stock is $y_E(x_1, s) = F_0(v(M) - c\,x_1 + s)$, decreasing in s. Writing a_E for the induced activity, $c\,y_E = v(a_E) - v(M) + c\,x_1 - s$, so $c(x_1 - y_E) = v(M) - v(a_E) + s \ge 0$, with equality only if $s = 0$ and $a_E = M$, in which case $y_E = 0 < x_1$.

Formula (9). If $c\,x_1 \le s$, then $v(M) - \max\{s, c\,x_1\} = v(M) - s$ and (9) reduces to $\max\{u^M(x_1), v(M) - s\}$. If $c\,x_1 > s$, then $v(M) - \max\{s, c\,x_1\} = v(M) - c\,x_1$ and $u^M(x_1) - (v(M) - c\,x_1) = -\Delta - c\,\tilde{x}(1 - x_1) < 0$, so (9) reduces to $v(M) - c\,x_1$. □

B.7 Proof of Lemma 3

Let I be an open interval containing x_1^* on which $W(x; s) = \alpha - c\,e\,x$. For activities whose disclosure lies in I, the user's period-1 objective equals $v(a) - \lambda(1 + \beta_u e)\sigma_0^2\,\xi(a, \gamma_1; 0) + \beta_u \alpha$, the static objective with cost $\Lambda_e := \lambda(1 + \beta_u e)$. Since x_1^* is interior to I and $a_1 > 0$, the first-order condition (4) holds: $x_1^* = \phi(a_1)/\sqrt{\Lambda_e}$. If $a_1 \neq \tilde{a}$, a small change of protection that moves the solution of (4) toward $\tilde{a}$ keeps the user's best response in I (by the maximum theorem, because her best response at γ_1 is unique by assumption) and raises her disclosure, which contradicts the assumption that the incumbent maximizes disclosure. Hence $a_1 = \tilde{a}$, $x_1^* = \phi^*/\sqrt{\Lambda_e} = \tilde{x}/\sqrt{1 + \beta_u e}$, and $\gamma_1 = \Gamma_{\Lambda_e}(\tilde{a}; \sigma_0^2) = \hat{\gamma}(\sigma_0^2, \Lambda_e)$. □

B.8 Proof of Theorem 1

Throughout, $\Lambda_P := \lambda(1 + \beta_u)$, $\Lambda_M := \lambda(1 + \beta_u \kappa)$, $x_k := s/c$, and $x_c(s) := (s - s_2)/(c\kappa)$. All static problems have prior variance σ_0^2, and we write $S_\Lambda(\gamma)$ and $a_\Lambda(\gamma)$ for $S_\Lambda(\gamma; \sigma_0^2)$ and the corresponding activity.

Step 0: the user's problem as a combination of static problems. Fix γ_1. By Lemma B.3, we may restrict the user to $a \in [0, a_\lambda(\gamma_1)]$, on which disclosure is at most $\tilde{x}$ and W is given by (9). Define

$$
\begin{aligned}
G_M(a) &:= v(a) - \Lambda_M \sigma_0^2\,\xi + \beta_u\big(v(\tilde{a}) - c\,\tilde{x}\big) && \text{(guarded)},\\
G_I(a) &:= v(a) - \lambda \sigma_0^2\,\xi + \beta_u\big(v(M) - s\big) && \text{(insured)},\\
G_P(a) &:= v(a) - \Lambda_P \sigma_0^2\,\xi + \beta_u\, v(M) && \text{(pegged)},
\end{aligned}
$$

with $\xi = \xi(a, \gamma_1; 0)$. Substituting (9), the user's objective is

$$
V(a) = v(a) - c\,\xi + \beta_u W(\xi) = \max\big\{G_M(a),\ \min\{G_I(a), G_P(a)\}\big\}. \tag{B.1}
$$

Each G is a static objective, maximized by $a = 0$ or by $a_\Lambda(\gamma_1)$ for the corresponding Λ (Lemma 1). Since $a_{\Lambda_P}(\gamma_1) < a_{\Lambda_M}(\gamma_1) < a_\lambda(\gamma_1)$, all these maximizers lie in $[0, a_\lambda(\gamma_1)]$, so the restriction is without loss. Note that $G_I \le G_P$ if and only if $c\,\xi \le s$.

Case A: $s \le s_2$. For every a, $G_M(a) - G_I(a) = \beta_u[u^M(\xi) - v(M) + s] \le \beta_u[u^M(0) - v(M) + s] = \beta_u(s - s_2) \le 0$ and $G_M(a) - G_P(a) = \beta_u[u^M(\xi) - v(M) + c\,\xi] = -\beta_u[\Delta + c\,\tilde{x}(1-\xi)] < 0$. Hence $V = \min\{G_I, G_P\}$.

A1 (upper bound). Let a^* be a best response with disclosure x^*. If $x^* > x_k$, then $V = G_P$ near a^*, so $a^* > 0$ is an interior local maximizer of G_P and satisfies (4) with $\Lambda = \Lambda_P$: $x^* = \phi(a^*)/\sqrt{\Lambda_P} \le x^P$. If $x^* \le x_k$, then $x^* \le \min\{x_k, \tilde{x}\}$ by Lemma B.3. Hence $x^* \le \bar{x}(s) := \max\{\min\{x_k, \tilde{x}\}, x^P\}$.

A2 (implementation). (a) Let $x \in (0, \min\{x_k, \tilde{x}\}]$, let $a^\circ \in [\tilde{a}, M)$ solve $\phi(a^\circ) = \sqrt{\lambda}\,x$, and let $\gamma^\circ = \Gamma_\lambda(a^\circ; \sigma_0^2) \ge 0$. By Lemma 1 and the participation inequality in the proof of Proposition 1, a° maximizes G_I. Because $c\,x \le s$, $V(a^\circ) = G_I(a^\circ)$, and $V \le G_I$ everywhere. Hence a° is a best response, and it is the only one, since any other best response would also maximize G_I and $G_I(0) < G_I(a^\circ)$. (b) If $x_k < x^P$ and $x \in (x_k, x^P]$, let $a^\circ \in [\tilde{a}, M)$ solve $\phi(a^\circ) = \sqrt{\Lambda_P}\,x$ and $\gamma^\circ = \Gamma_{\Lambda_P}(a^\circ; \sigma_0^2) \ge 0$. Then a° maximizes G_P, because $G_P(a^\circ) - G_P(0) = v(a^\circ) - c(1+\beta_u)x \ge v(\tilde{a}) - \sqrt{1+\beta_u}\,c\,\tilde{x} \ge 0$ by Assumption 2(c); moreover $V(a^\circ) = G_P(a^\circ)$ since $c\,x > s$, and $V \le G_P$. So x is implementable. (c) $x = 0$ is implemented by $\gamma = \infty$.

A3 (the incumbent's choice). In Case A, $u^M(x) \le v(M) - s$ for all x, so by Lemma 2 the incumbent's period-2 stock is $y_A(x, s)$ if $x \le x_k$ and 0 otherwise. Its objective $J(x) = \sigma_0^2[x + \beta_p y_A(x, s)]$ for $x \le x_k$, and $J(x) = \sigma_0^2 x$ for $x > x_k$, is increasing on each piece.

- If $s \ge s_1$, then $\bar{x}(s) = \tilde{x} \le x_k$, so the incumbent implements $\tilde{x}$, which by A2(a) requires $a^\circ = \tilde{a}$ and hence $\gamma_1 = \Gamma_\lambda(\tilde{a}; \sigma_0^2) = \tilde{\gamma}$. This is regime (iii) for $s \in [s_1, s_2]$.
- If $c\,x^P \le s < s_1$, then $\bar{x}(s) = x_k$ and the incumbent implements x_k, earning $\sigma_0^2(1 + \beta_p)x_k$ because $y_A(x_k, s) = x_k$. All protection levels that implement x_k give it the same payoff, and by (E2) it chooses the one with the highest activity. At a best response a_1 with disclosure x_k, optimality from the left requires $V'(a_1^-) = G_I'(a_1) \ge 0$, i.e., $x_k \le \phi(a_1)/\sqrt{\lambda}$. The largest such a_1 is $a_1 = \phi^{-1}(\sqrt{\lambda}\,x_k)$ on the decreasing branch, and it is implementable by A2(a).
- If $s < c\,x^P$, then $\bar{x}(s) = x^P > x_k$. The best implementable disclosure with retention is x_k, worth $\sigma_0^2(1 + \beta_p)x_k$, and the best without retention is x^P, worth $\sigma_0^2 x^P$. The incumbent retains the user if and only if $(1 + \beta_p)s/c \ge x^P$, i.e., $s \ge s_P$; at $s = s_P$, rule (E2) selects retention, which induces the higher activity. For $s < s_P$ the incumbent implements x^P with $\gamma_1 = \Gamma_{\Lambda_P}(\tilde{a}; \sigma_0^2) = \hat{\gamma}(\sigma_0^2, \Lambda_P)$ and $a_1 = \tilde{a}$.

The period-2 outcomes follow from Lemma 2. In regime (ii), $x_1^* = s/c$ makes the incumbent's best offer equal to the rival's, so it serves the user with full protection and collects no new data. This proves (i)–(iii) for $s \le s_2$.

Case B: $s > s_2$. Then $x_k > s_2/c > \tilde{x}$, so $G_I < G_P$ on the relevant domain and $V = \max\{G_M, G_I\}$. Since the maximum of a pointwise maximum is the larger of the

two maxima, the user chooses the insured response $a_\lambda(\gamma_1)$ if

$$S_\lambda(\gamma_1) + \beta_u(v(M) - s) \;\geq\; S_{\Lambda_M}(\gamma_1) + \beta_u\big(v(\tilde{a}) - c\,\tilde{x}\big) \quad\Longleftrightarrow\quad R(\gamma_1) \geq \beta_u(s - s_2),$$

and a maximizer of G_M otherwise. Ties are broken toward the insured response, which the incumbent prefers (see below). By Lemma 2, for $x \leq \tilde{x} < x_k$ the incumbent always retains the user. Its period-2 stock is $\tilde{x} + \kappa x$ for $x \leq x_c(s)$ and $y_A(x, s)$ for $x > x_c(s)$; the two expressions agree at $x_c(s)$ and both are increasing. Hence the incumbent's objective is increasing in x_1, and it maximizes period-1 disclosure.

Properties of R. Let $\gamma_M := \hat{\gamma}(\sigma_0^2, \Lambda_M) > \tilde{\gamma}$.

(R1) $R \geq 0$ is continuous and nondecreasing on $[0, \tilde{\gamma}]$. By Lemma 1, the one-sided derivatives of R are of the form $\phi(a_\lambda)^2 - \phi(a_{\Lambda_M})^2$, with the convention that a term is zero when the corresponding optimum is $a = 0$ (and $R = 0$ when the insured optimum is $a = 0$, since then $S_{\Lambda_M} = S_\lambda = 0$). For $\gamma \leq \tilde{\gamma}$ they are nonnegative, because $a_{\Lambda_M}(\gamma) < a_\lambda(\gamma) \leq \tilde{a}$ and ϕ is increasing on $[0, \tilde{a}]$.

(R2) On $[\tilde{\gamma}, \gamma_M]$, R is concave. There, $a_\lambda(\gamma) \geq \tilde{a}$, so $S'_\lambda = \phi(a_\lambda(\gamma))^2$ is nonincreasing. The interior branch of S_{Λ_M} has derivative $\phi(a_{\Lambda_M}(\gamma))^2$, nondecreasing because $a_{\Lambda_M}(\gamma) \leq \tilde{a}$, so S_{Λ_M}, as the maximum of 0 and a convex function, is convex.

(R3) On $[\gamma_M, \infty)$, R is strictly decreasing: $a_\lambda(\gamma) > a_{\Lambda_M}(\gamma) \geq \tilde{a}$, so $R' = \phi(a_\lambda)^2 - \phi(a_{\Lambda_M})^2 < 0$ (the guarded optimum is interior by Assumption 2(c)).

(R4) For $\gamma \geq \tilde{\gamma}$, $R(\gamma) < \beta_u \kappa\, c\, x_\lambda(\gamma)$. Evaluating the guarded objective at $a_\lambda(\gamma)$, which is not one of its maximizers, gives $S_{\Lambda_M}(\gamma) > S_\lambda(\gamma) - (\Lambda_M - \lambda)\sigma_0^2 x_\lambda(\gamma) = S_\lambda(\gamma) - \beta_u \kappa\, c\, x_\lambda(\gamma)$.

(R5) $R(\gamma_M) > \beta_u \kappa\, c\, x^M$. Evaluating the insured objective at $\tilde{a}$, which is not its maximizer at γ_M, gives $S_\lambda(\gamma_M) > v(\tilde{a}) - c\, x^M = S_{\Lambda_M}(\gamma_M) + \beta_u \kappa\, c\, x^M$.

By (R2)–(R3), R restricted to $[\tilde{\gamma}, \infty)$ is single-peaked, with its maximum at some $\gamma_R \in (\tilde{\gamma}, \gamma_M)$. It is strictly increasing just to the right of $\tilde{\gamma}$ because $R'(\tilde{\gamma}^+) \geq \phi^{*2} - \phi(a_{\Lambda_M}(\tilde{\gamma}))^2 > 0$, and strictly decreasing to the left of γ_M because $R'(\gamma_M^-) \leq \phi(a_\lambda(\gamma_M))^2 - \phi^{*2} < 0$. Also, $R(\tilde{\gamma}) > 0$, so $\bar{s} > s_2$: $S_\lambda(\tilde{\gamma}) = v(\tilde{a}) - c\,\tilde{x} > 0$, and either the guarded optimum at $\tilde{\gamma}$ is $a = 0$, so that $S_{\Lambda_M}(\tilde{\gamma}) = 0$, or it is some $a > 0$ with positive disclosure, at which the insured objective is strictly larger than the guarded one. Moreover, $\hat{s} = s_2 + R(\gamma_R)/\beta_u > \bar{s}$; and, by (R4), $R(\gamma_R) < \beta_u \kappa c\, \tilde{x}$, so $\hat{s} < s_3$.

The incumbent's choice. The insured disclosure $x_\lambda(\gamma)$ is strictly decreasing on $[\tilde{\gamma}, \infty)$ and equals $\tilde{x}$ at $\tilde{\gamma}$. Protection below $\tilde{\gamma}$ is never useful: by (R1), if some $\gamma < \tilde{\gamma}$ satisfies $R(\gamma) \geq \beta_u(s - s_2)$, so does $\tilde{\gamma}$, which yields the larger disclosure $\tilde{x}$. Guarded responses disclose at most $x^M < x_\lambda(\gamma)$ for all $\gamma \in [\tilde{\gamma}, \gamma_M]$.

- If $s_2 < s \leq \bar{s}$, then $R(\tilde{\gamma}) \geq \beta_u(s - s_2)$, so the incumbent offers $\tilde{\gamma}$ and obtains $x_1^* = \tilde{x}$: regime (iii).
- If $\bar{s} < s \leq \hat{s}$, the set $\{\gamma \geq \tilde{\gamma} : R(\gamma) \geq \beta_u(s - s_2)\}$ is a nonempty interval whose smallest element is $\gamma^*(s) \in (\tilde{\gamma}, \gamma_R]$, and the incumbent offers $\gamma^*(s)$. Since R is continuous and strictly increasing on $[\tilde{\gamma}, \gamma_R]$, $\gamma^*(s)$ is continuous and strictly increasing, so

$x_1^* = x_\lambda(\gamma^*(s))$ is continuous and strictly decreasing, with $x_1^* \geq x_\lambda(\gamma_R) > x_\lambda(\gamma_M) > \xi(\tilde{a}, \gamma_M; 0) = x^M$ (the last inequality because $a_\lambda(\gamma_M) > \tilde{a}$). Moreover, by (R4), $\beta_u(s - s_2) = R(\gamma^*(s)) < \beta_u \kappa c\, x_1^*$, i.e., $x_1^* > x_c(s)$, so the incumbent is constrained in period 2 and the user obtains $v(M) - s$: regime (iv).

- If $s > \hat{s}$, no protection level induces the insured response, and guarded disclosures are at most x^M, attained at γ_M, where the user engages by Assumption 2(c). Hence $x_1^* = x^M$. By (R5), $\beta_u(s - s_2) > R(\gamma_M) > \beta_u \kappa c\, x^M$, so $x^M < x_c(s)$: the incumbent is unconstrained in period 2 and the outcome coincides with Proposition 2: regime (v).

Ranking and protection. From the regimes, $\min_s x_1^* = s_P/c = x^P/(1+\beta_p)$ (attained at $s = s_P$), $\max_s x_1^* = \tilde{x}$, and $x^P < x^M$ because $\kappa < 1$. The jump at s_P is downward because $s_P/c < x^P$, and the jump above $\hat{s}$ is downward because $x_\lambda(\gamma_R) > x^M$. In regimes (i), (iii), and (v), $\gamma_1 = \hat{\gamma}(\sigma_0^2, \lambda(1+\beta_u e))$, which by Proposition 1(c) is decreasing in the implied disclosure. In regime (ii), $\gamma_1 = \sigma_0^2(c/s - 1) - 1/a_1(s)$ is decreasing in s. In regime (iv), $\gamma^*(s)$ is increasing in s. At the two jumps, protection jumps up: at s_P, disclosure falls and activity rises from $\tilde{a}$, and at $\hat{s}$ protection rises from γ_R to γ_M. □

B.9 Proof of Proposition 3

(a) and (b). By Theorem 1 and Lemma 2:

- on $[0, s_P)$, $U = v(\tilde{a}) - cx^P + \beta_u(v(M) - cx^P) = v(\tilde{a}) - \sqrt{1+\beta_u}\, c\, \tilde{x} + \beta_u v(M)$;
- on $[s_P, s_1]$, $U(s) = v(a_1(s)) - s + \beta_u(v(M) - s)$, which is strictly decreasing because $a_1(s) = \phi^{-1}(\sqrt{\lambda} s/c)$ is decreasing;
- on $[s_1, \bar{s}]$, $U(s) = v(\tilde{a}) - c\,\tilde{x} + \beta_u(v(M) - s)$, strictly decreasing and continuous with the previous piece at s_1;
- on $(\bar{s}, \hat{s}]$, the user is indifferent between the insured and the guarded response at $\gamma^*(s)$, so $U(s) = S_{\Lambda_M}(\gamma^*(s)) + \beta_u(v(\tilde{a}) - c\,\tilde{x})$;
- on $(\hat{s}, \infty)$, $U = U(\infty) = S_{\Lambda_M}(\gamma_M) + \beta_u(v(\tilde{a}) - c\,\tilde{x})$.

The jump at s_P is $v(a_1(s_P)) - v(\tilde{a}) + (1+\beta_u)(cx^P - s_P) > 0$. Since S_{Λ_M} is nondecreasing, strictly increasing wherever it is positive (its derivative is then $\phi(a_{\Lambda_M})^2 > 0$), and $S_{\Lambda_M}(\gamma_M) > 0$ by Assumption 2(c), the inequality $\gamma^*(s) \leq \gamma_R < \gamma_M$ gives $U(s) < U(\infty)$ on $(\bar{s}, \hat{s}]$. At $s = \bar{s}$, $R(\tilde{\gamma}) = \beta_u(\bar{s} - s_2)$ gives $U(\bar{s}) = S_{\Lambda_M}(\tilde{\gamma}) + \beta_u(v(\tilde{a}) - c\,\tilde{x}) < U(\infty)$. This proves (b). Finally, using $(1+\beta_u)s_P < (1+\beta_u)c\, x^P = \sqrt{1+\beta_u}\, c\, \tilde{x}$ and $v(a_1(s_P)) \geq v(\tilde{a})$,

$$U(s_P) - U(\infty) > \beta_u \Delta + c\,\tilde{x}\big(\sqrt{1+\beta_u\kappa} + \beta_u - \sqrt{1+\beta_u}\big) \geq \beta_u \Delta + c\,\tilde{x}\,\tfrac{\beta_u}{2} > 0,$$

because $\sqrt{1+\beta_u\kappa} \geq 1$ and $\sqrt{1+\beta_u} \leq 1 + \beta_u/2$. Combining the pieces, U is maximized at s_P.

(c). On $[0, s_P)$, $\Pi_A = \sigma_0^2 x^P$; on $[s_P, s_1]$, $\Pi_A = \sigma_0^2(1+\beta_p)s/c$, which is continuous at s_P by the definition of s_P and increasing; on $[s_1, \bar{s}]$, $\Pi_A = \sigma_0^2[\tilde{x} + \beta_p y_A(\tilde{x}, s)]$, which

is increasing in s (Lemma 2) and continuous at s_1 because $y_A(\tilde{x}, s_1) = \tilde{x}$. At $s = \hat{s}$, $x_1^* > x_c(\hat{s})$, and since $y_A(\cdot, \hat{s})$ is increasing with $y_A(x_c(\hat{s}), \hat{s}) = \tilde{x} + \kappa x_c(\hat{s})$,

$$\Pi_A(\hat{s}) > \sigma_0^2\left[x_c(\hat{s}) + \beta_p(\tilde{x} + \kappa x_c(\hat{s}))\right] \geq \sigma_0^2\left[x^M + \beta_p(\tilde{x} + \kappa x^M)\right] = \Pi_A(\infty),$$

where the second inequality uses $x_c(\hat{s}) \geq x^M$: for all $s > \hat{s}$ we showed $x_c(s) > x^M$, and x_c is continuous. □

B.10 Proof of Proposition 4

With the right, the user erases the incumbent's data at the start of period 2, because erasure raises her payoff from every incumbent offer and does not affect the rival's offers. The period-2 subgame is then the one of Lemma 2 with $x_1 = 0$: the incumbent serves, the user obtains $W^R := \max\{v(\tilde{a}) - c\,\tilde{x}, v(M) - s\}$, and the incumbent's period-2 stock does not depend on x_1. In period 1 the user's continuation is constant, so she solves the static problem with $\Lambda = \lambda$, and the incumbent maximizes disclosure. Proposition 1 gives $\tilde{\gamma}$, $\tilde{a}$, and $\tilde{x}$. Hence $U^R(s) = v(\tilde{a}) - c\,\tilde{x} + \beta_u W^R$. Comparing with the proof of Proposition 3:

- for $s < s_P$ (poaching): $U^R - U = c\,\tilde{x}(\sqrt{1+\beta_u} - 1) - \beta_u s$, positive if and only if $s < s_R$;
- for $s \in [s_P, s_1)$: $U - U^R = v(a_1(s)) - v(\tilde{a}) + c\,\tilde{x} - s > 0$;
- for $s \in [s_1, s_2]$: $U^R = U$;
- for $s \in (s_2, \bar{s}]$: $U^R - U = \beta_u(s - s_2) > 0$;
- for $s > \bar{s}$: $U^R = (1+\beta_u)(v(\tilde{a}) - c\,\tilde{x})$ and $U(s) \leq U(\infty) = v(\tilde{a}) - \sqrt{1+\beta_u\kappa}\,c\,\tilde{x} + \beta_u(v(\tilde{a}) - c\,\tilde{x})$, so $U^R - U \geq c\,\tilde{x}(\sqrt{1+\beta_u\kappa} - 1) > 0$. □

B.11 Proof of Proposition 5

Call the platform the user chooses in period 1 the incumbent. For $x_1 \leq \tilde{x}$, $c\,x_1 \leq c\,\tilde{x} = s_1 \leq s$, so by Lemma 2 the incumbent serves the user in period 2. The other platform therefore earns nothing in either period, and the user's continuation is $W(x_1; s)$. Let $\Pi(x_1)$ denote the incumbent's period-2 stock. In particular $\Pi(0) > 0$: with $x_1 = 0$ the incumbent must give the user $\max\{v(\tilde{a}) - c\,\tilde{x}, v(M) - s\} < v(M)$, which it can do while collecting a positive stock. The user's lifetime payoff from a platform offering γ in period 1 is $U(\gamma) := \max_a\{v(a) - c\,\xi(a, \gamma; 0) + \beta_u W(\xi(a, \gamma; 0))\}$. Since W is nonincreasing and $v \leq v(M)$, $U(\gamma) < U(\infty) = v(M) + \beta_u W(0)$ for every $\gamma < \infty$: at finite γ the user either engages at a level below M or discloses a positive amount.

Suppose that in some equilibrium the user chooses a platform whose offer is $\gamma < \infty$. The other platform earns zero. By offering $\gamma = \infty$ instead, it would attract the user and earn $\beta_p\sigma_0^2\Pi(0) > 0$, a profitable deviation. Hence in every equilibrium the user's period-1 platform offers $\gamma = \infty$, so $x_1^* = 0$. Conversely, both platforms offering $\gamma = \infty$ is an equilibrium: a platform that deviates to $\gamma < \infty$ loses the user. The period-2 outcome

follows from Lemma 2 with $x_1 = 0$. If $s \geq s_2$, then $u^M(0) = v(\tilde{a}) - c\,\tilde{x} \geq v(M) - s$, so the incumbent is unconstrained and learns $\tilde{x}$. □

C Numerical example

The figures use $v(a) = \frac{K}{2M} a(2M - a)$ with $M = 1$ and $K = 0.1$, $\lambda = 1$, $\sigma_0^2 = 0.25$ (so $c = 0.25$), and $\beta_u = \beta_p = 0.9$. Then $\tilde{a} = 2/3$, $\tilde{x} \approx 0.1217$, $\kappa \approx 0.878$, $x^P \approx 0.0883$, $x^M \approx 0.0910$, and $\Delta \approx 0.0056$. Assumption 2(b) holds with slack ≈ 0.007 and (c) with slack ≈ 0.0025. The thresholds are $s_P \approx 0.0116$, $s_1 \approx 0.0304$, $s_2 \approx 0.0360$, $\bar{s} \approx 0.0516$, $\hat{s} \approx 0.0582$, and $s_3 \approx 0.0627$. At $\tilde{\gamma}$ the guarded user would not be active at all ($S_{\Lambda_M}(\tilde{\gamma}) = 0$), so $\bar{s} = s_2 + (v(\tilde{a}) - c\,\tilde{x})/\beta_u$. In the transition regime, except a very small interval at the top ($s \in (0.05823, \hat{s}]$), the binding guarded option is not being active at all, and the user's payoff is $\beta_u(v(\tilde{a}) - c\,\tilde{x}) \approx 0.0126$.

We computed the equilibrium in two independent ways. First, we used the closed forms in Theorem 1, and for the transition regime we solved $R(\gamma) = \beta_u(s - s_2)$ numerically. Second, we wrote a brute-force solver that (i) computes the Bertrand outcome in period 2 by searching over the protection levels and activity levels of both platforms, and (ii) in period 1, computes the user's global best response to each protection level on a fine grid and then lets the incumbent choose the best protection level. The two methods give the same results up to the grid precision (the largest difference in x_1^* is below 10^{-4}), for the parameters above and for several random parameter values that satisfy Assumption 2. In all these cases, the user's payoff is highest at s_P and the incumbent's profit is highest in the transition regime.